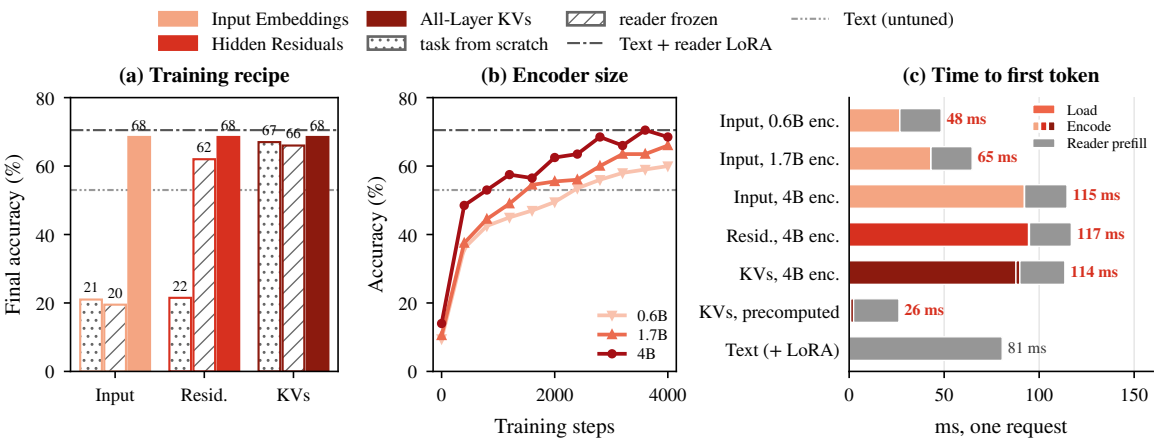

Input Embeddings
Hidden Residuals
All-Layer KVs
task from scratch
reader frozen
Text + reader LoRA
Text (untuned)
(a) Training recipe
Final accuracy (%)
80
60
40
20
0
68
21
20
62
68
22
67
66
68
Input
Resid.
KVs
(b) Encoder size
Accuracy (%)
80
60
40
20
0
0
2000
4000
Training steps
0.6B
1.7B
4B
(c) Time to first token
Load
Encode
Reader prefill
Input, 0.6B enc.
48 ms
Input, 1.7B enc.
65 ms
Input, 4B enc.
115 ms
Resid., 4B enc.
117 ms
KVs, 4B enc.
114 ms
KVs, precomputed
26 ms
Text (+ LoRA)
81 ms
0
50
100
150
ms, one request